BIG DATA ANALYTICS PROJECT ON **GOOGLE TRENDS** +**WORLD BANK INDICATORS**

From Spike to Signals

What the world searches for, and how it spreads

Google Trends!

Google Trends analyzes the popularity of search queries over time and across regions.

Top Terms: "Baseline Popularity" - Consistently searched queries (**Long-term interest**).

Rising Terms: "Momentum" - Queries with sudden growth spikes (**Short-term events**).

Key Metrics:

Score (0-100): Relative search volume normalized by region/time (**Used for Top Terms**).

Percent Gain: Growth rate compared to the previous period (**Used for Rising Terms**).

Project Introduction

Business Problem:

Spikes ≠ Signals. High volatility makes cross-country comparison difficult

Lack of structural context (Why do trends fade fast in some markets?

Project Goal: "From Spikes to Signals"

Build a country-level **diagnostic framework**.

Explain search behaviors using **Macro Indicators** (GDP, Digital Adoption, etc).

Data Source & Methodology

Dataset Overview:

Coverage:
42 Countries, **Aug-Oct 2025** Refreshments, **2020-2025** historical window.

Volume:
10M+ rows, **300k+** unique Terms.

Context:
Integrated with **World Bank WDI** (GDP, Internet Usage, Freedom of Speech index etc.).

Methodology & Cleaning:

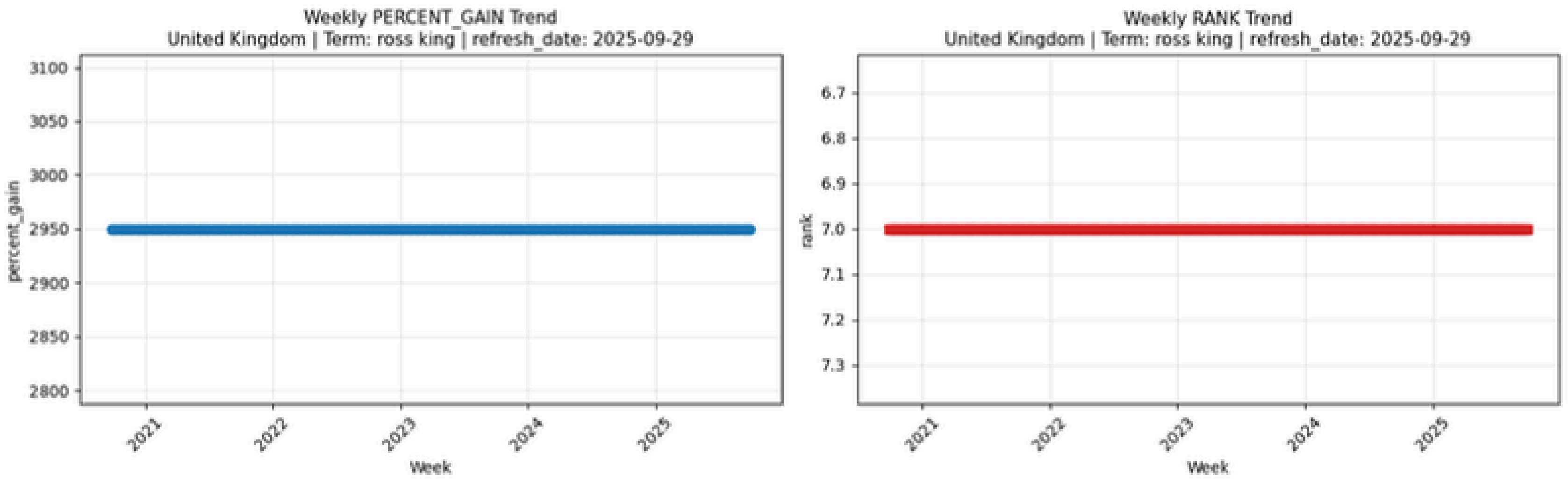
Rising Terms: Drop Week -> Compress to Unique (Country-Term) Events.

Top Terms (Dual-Track):
Full Data-> Analyze Refresh Patterns.
Static Snapshot -> Analyze Score Correlati

Rising Term Data Structure:

Under the same refresh_date, **Constant Value** for both Rank & Percent_gain.

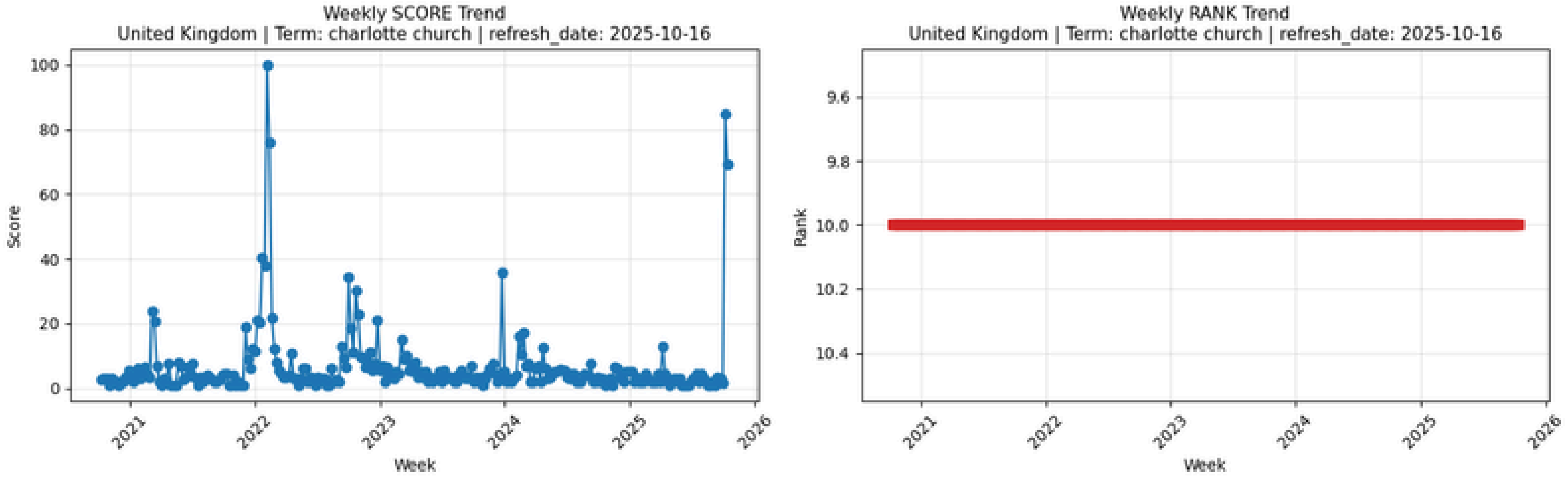
Picture 1 - Understand the Rising Term Dataset Refreshment Mechanism



Top Term Data Structure:

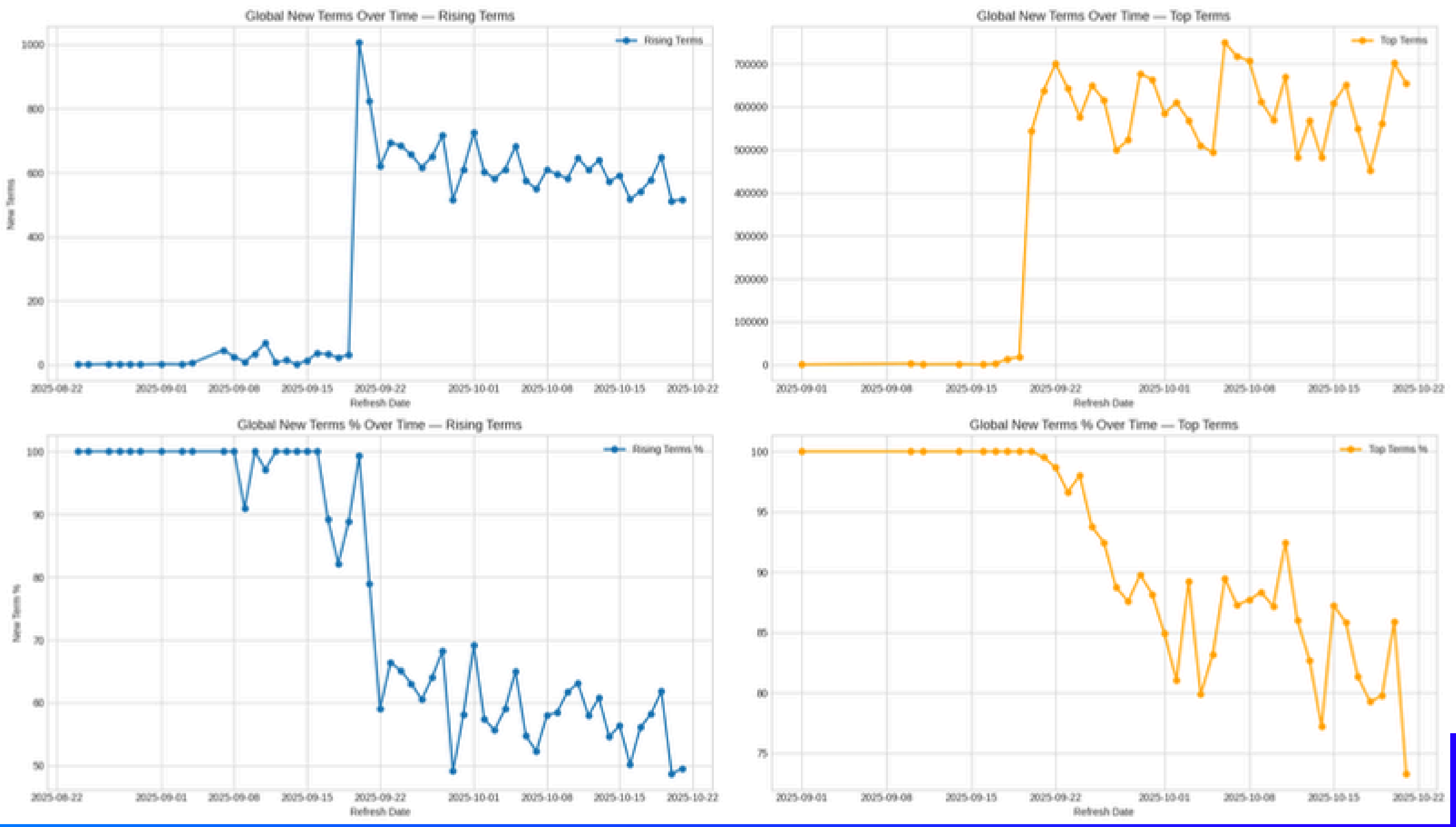
Under the same refresh_date, **Constant Value** for Rank, but 100 **actual historical data** (among 5 years) for Score.

Picture 2 - Understand the Top Term Dataset Refreshment Mechanism



How Search Energy Moves Over Time

Picture 3 - Google Trends Terms Update Dashboard
(New Term Volume & Rate Over Time)



NEW

Nearly **2,620** new terms replace old ones daily, achieving almost **70%** new term churn rate.

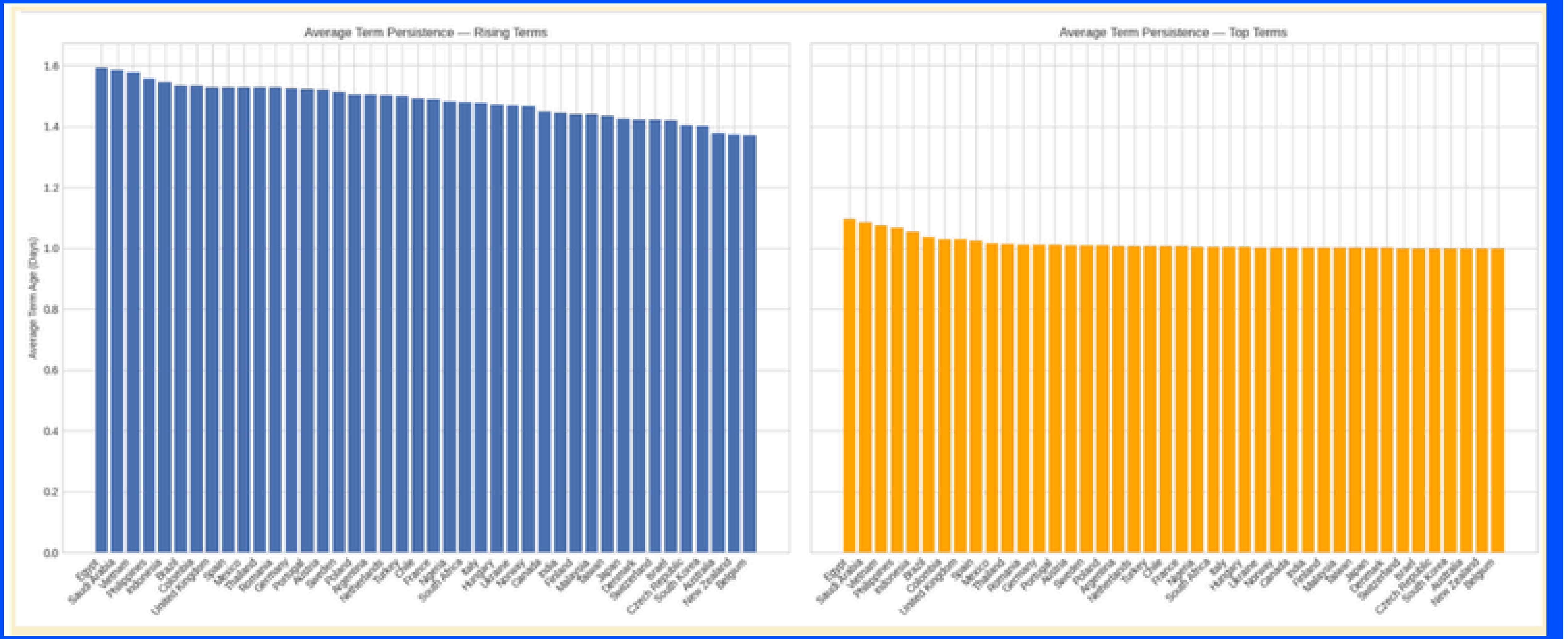
Rising terms **fluctuate** greatly, yet maintain a **high proportion** of one-off daily entries.

In **mid-September 2025**, Google's update behavior changes, **increasing** new Rising terms significantly.

Implication:

Rising is **More Stable,** Top terms shows **more churn rate %**

Why Most Trends Die Young



Country Averages

- When we aggregate **term_age** by **country**, most countries cluster just above 1 day for both **Rising** and **Top**.
- A few countries show slightly **longer** average **ages**, but even at the high end average Rising term age remains close to **one** or **two** days.

Rising Terms (Distinct)

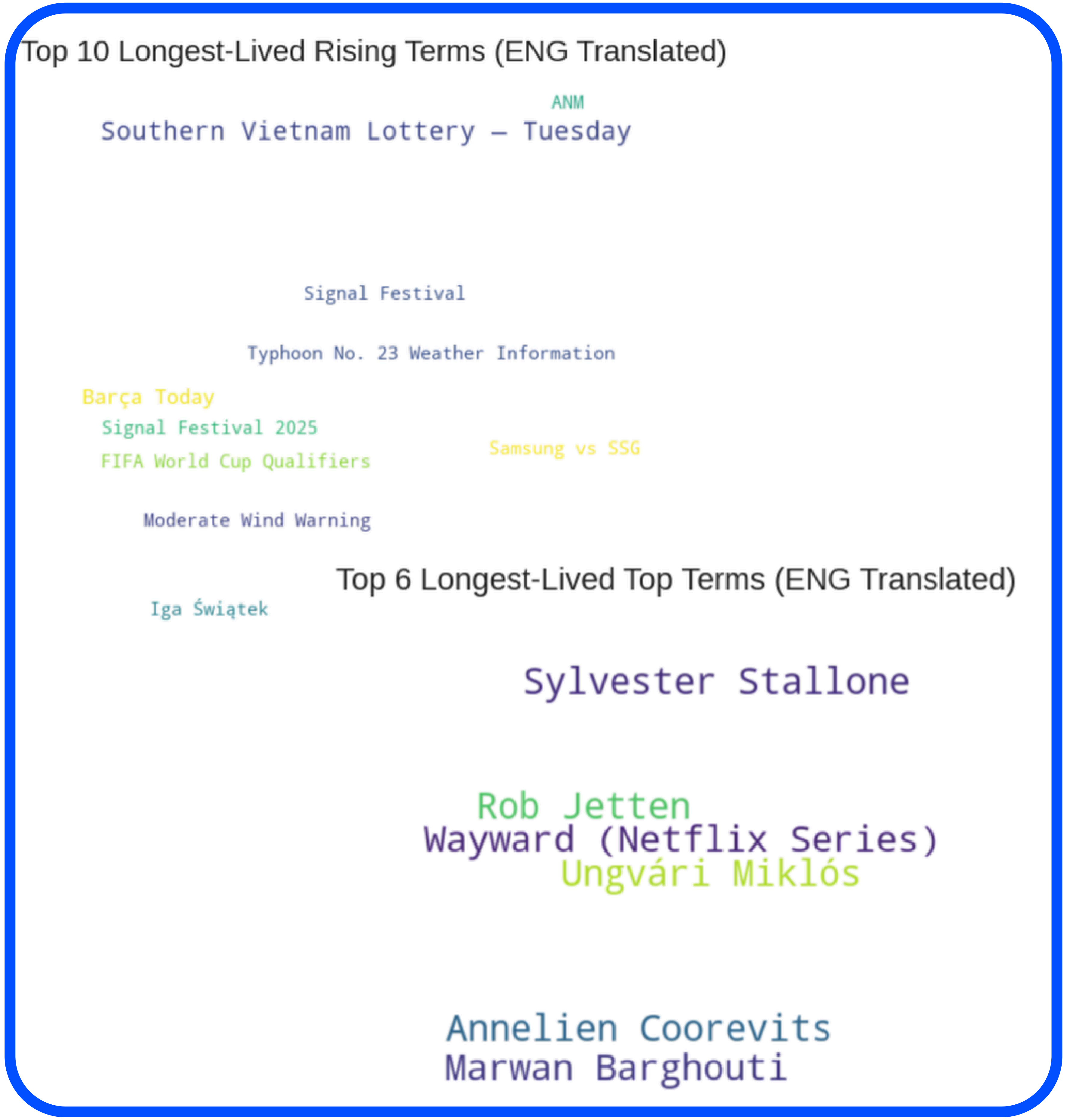
- With age 1 day: **8,637**.
- Age 2 days: **6,804**.
- Age 3 days: **217**; age 4 days: **8**; age 5 days: **1**; age 8 days: **1**.
- In total, roughly **98%** of Rising terms in our panel survive at most two days before being replaced.

Top Terms (Distinct)

- With age 1 day: **21,052**.
- Age 2 days: **334**; age 3 days: **25**; age 4 days: **5**; age 5 days: **1**.
- More than **98%** of Top terms in the panel also appear for only one day at that specific rank position, reflecting the daily refresh logic.

Trending is short-lived; focus on long-term trends.

Content Pulse via Word Clouds



Rising Terms Content Pulse:

Sports events, **matchups**, **tournaments**, **news**, **entertainment.**

Short **tournaments** feature repeated **match-ups** frequently.

Top Terms Content Pulse:

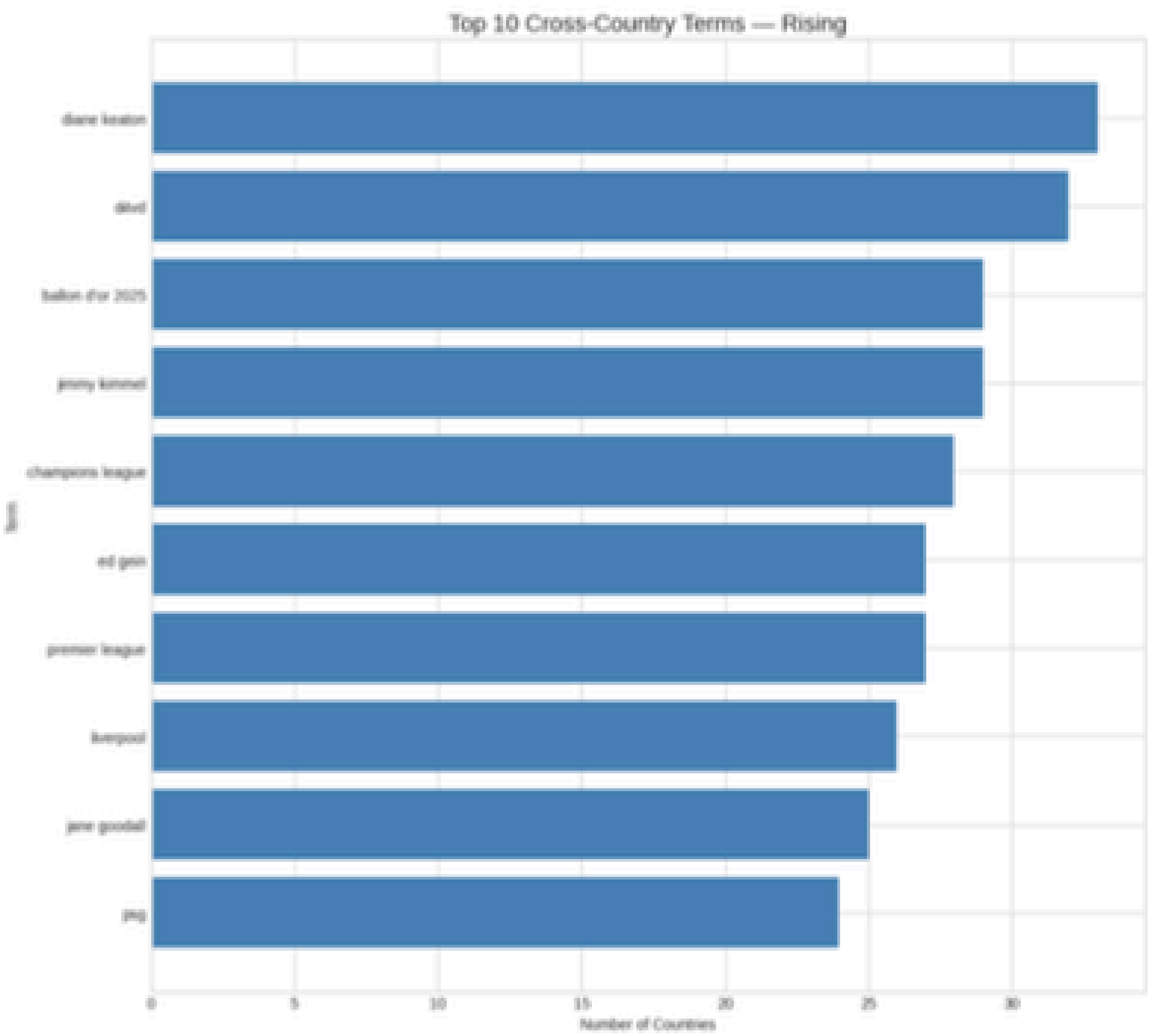
Word cloud highlights **clubs**, **competitions**, **celebrities, tv show.**

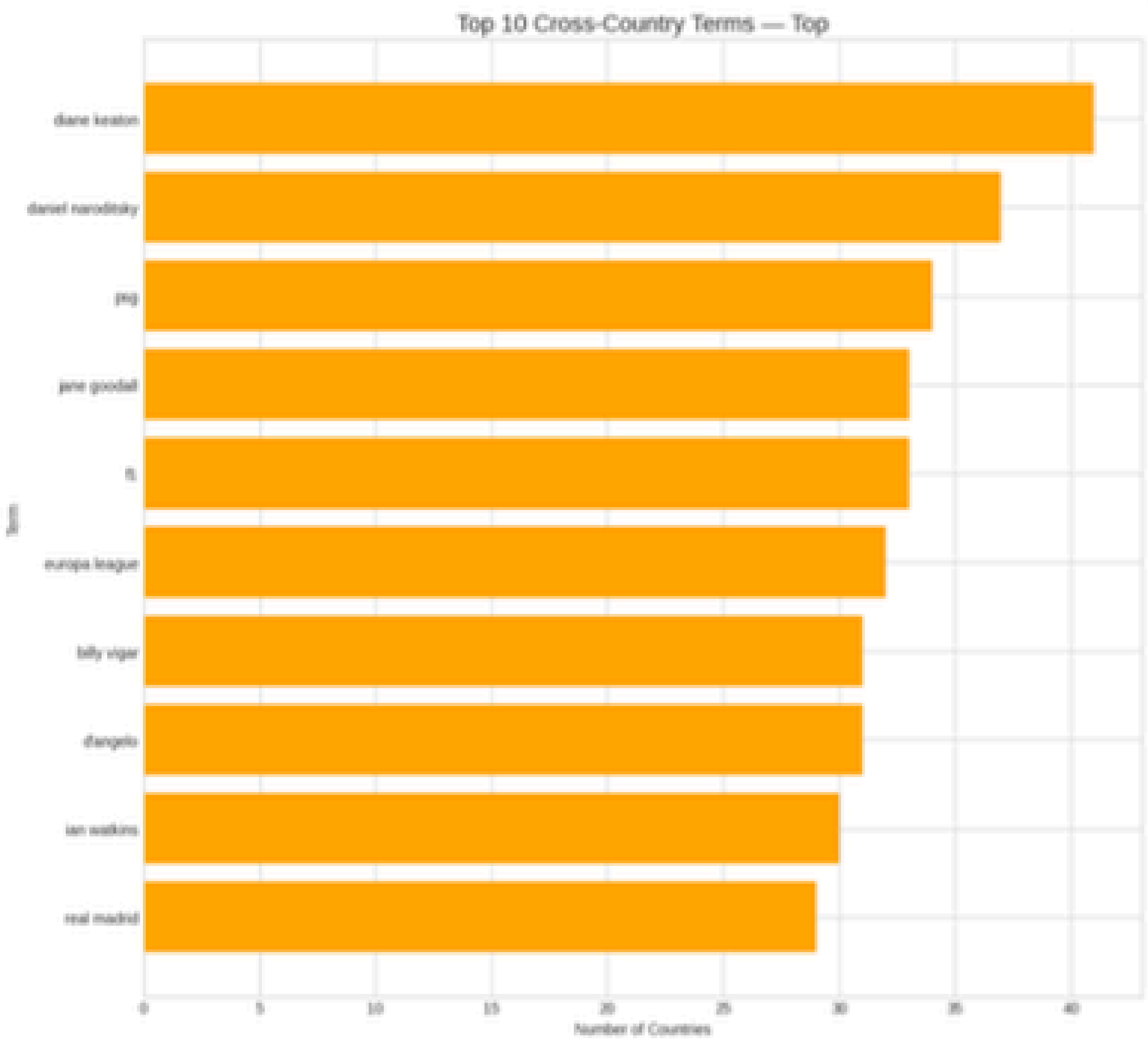
Terms fluctuate in **rank**, rarely disappearing completely.

Implication:

Rising captures **spikes**; top captures **background**. Integrating both layers is essential for **modeling**.

Cross-Country Term Diffusion





Observed Pattern

Rising terms capture **short-term regional spikes**, often driven by entertainment, sports events, or public figures.

Top terms reflect **persistent global culture**, dominated by major sports teams and widely recognized personalities.

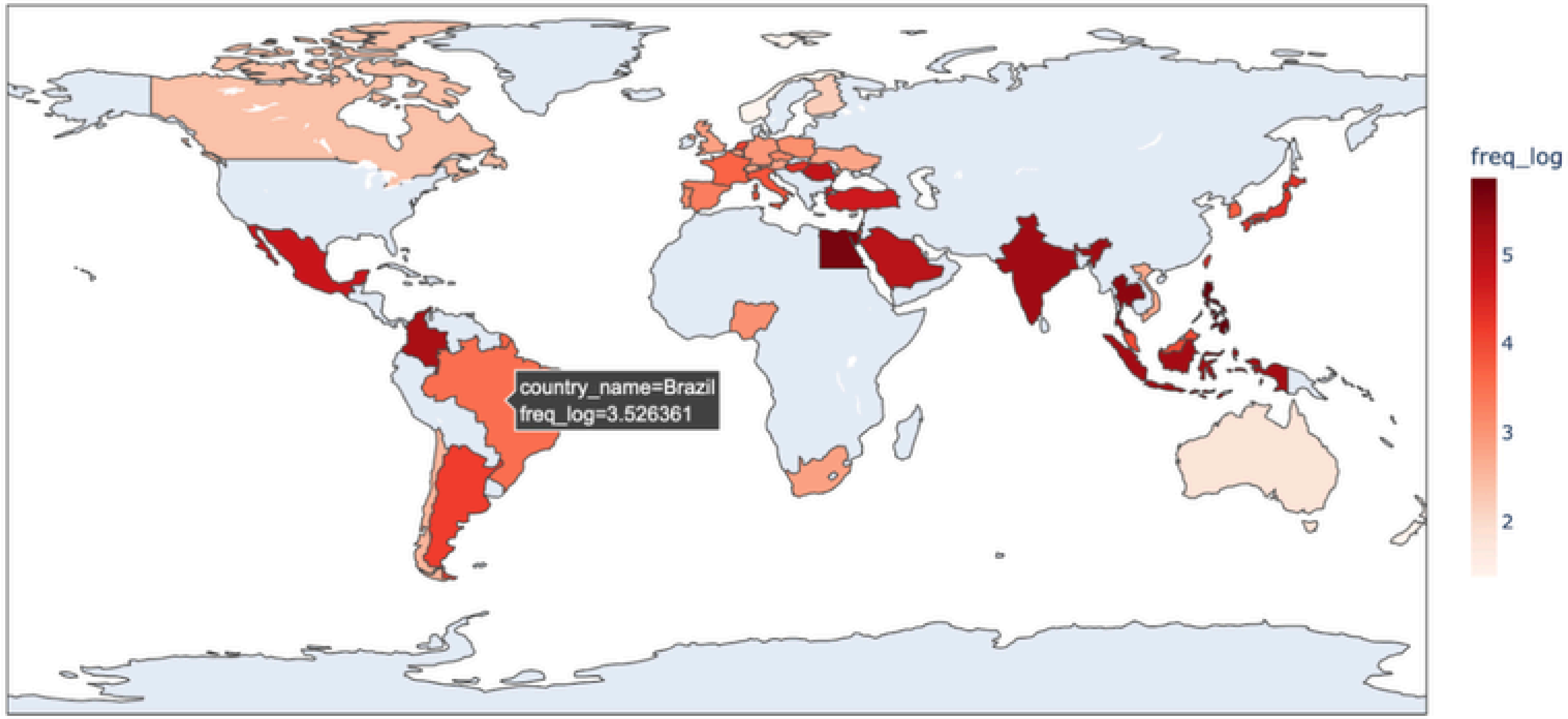
Gambling-related terms also appear in this diffusion pattern before, hinting at underlying spam or synthetic activity that varies by region.

Interpretation

Global Top terms form a **shared culture**. Rising terms show **local shocks**, which may transition into the background layer, informing our modeling.

Spam Attack Spatial Diffusion Map

PGLucky88 Spam — Global Spread (Log Frequency)

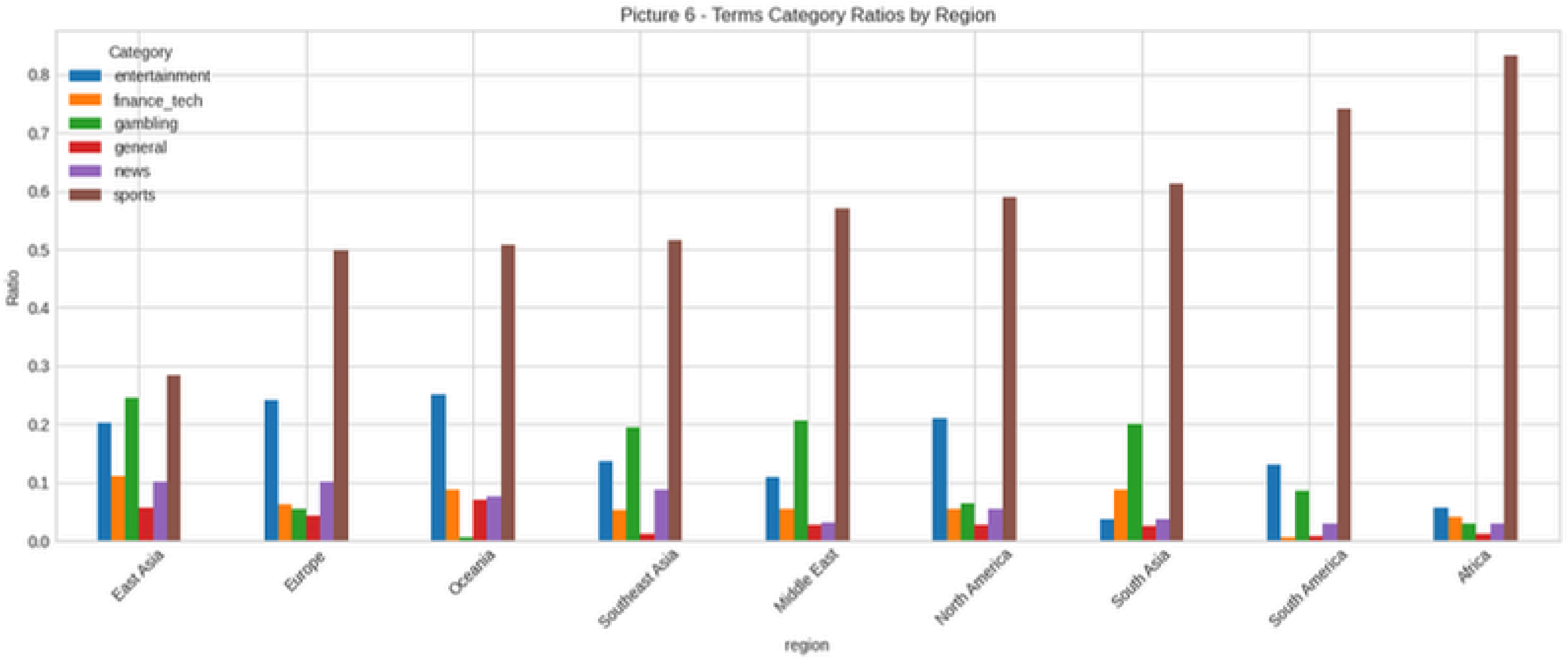


- East & Southeast Asia (Philippines, Indonesia, Malaysia, Vietnam) show the strongest concentrations, consistent with known regional spam networks.
- South Asia and parts of the Middle East also display notable activity.

*This map Indicates **coordinated gambling**-related spam networks **targeting** specific countries.*
*It Highlights the importance of **spam detection** to avoid distorted **cross-country comparisons**.*

- Gambling-related spikes do not reflect genuine organic interest.
- Their cross-country patterns align with coordinated spam behavior rather than natural cultural diffusion.

Category Distribution Across Regions



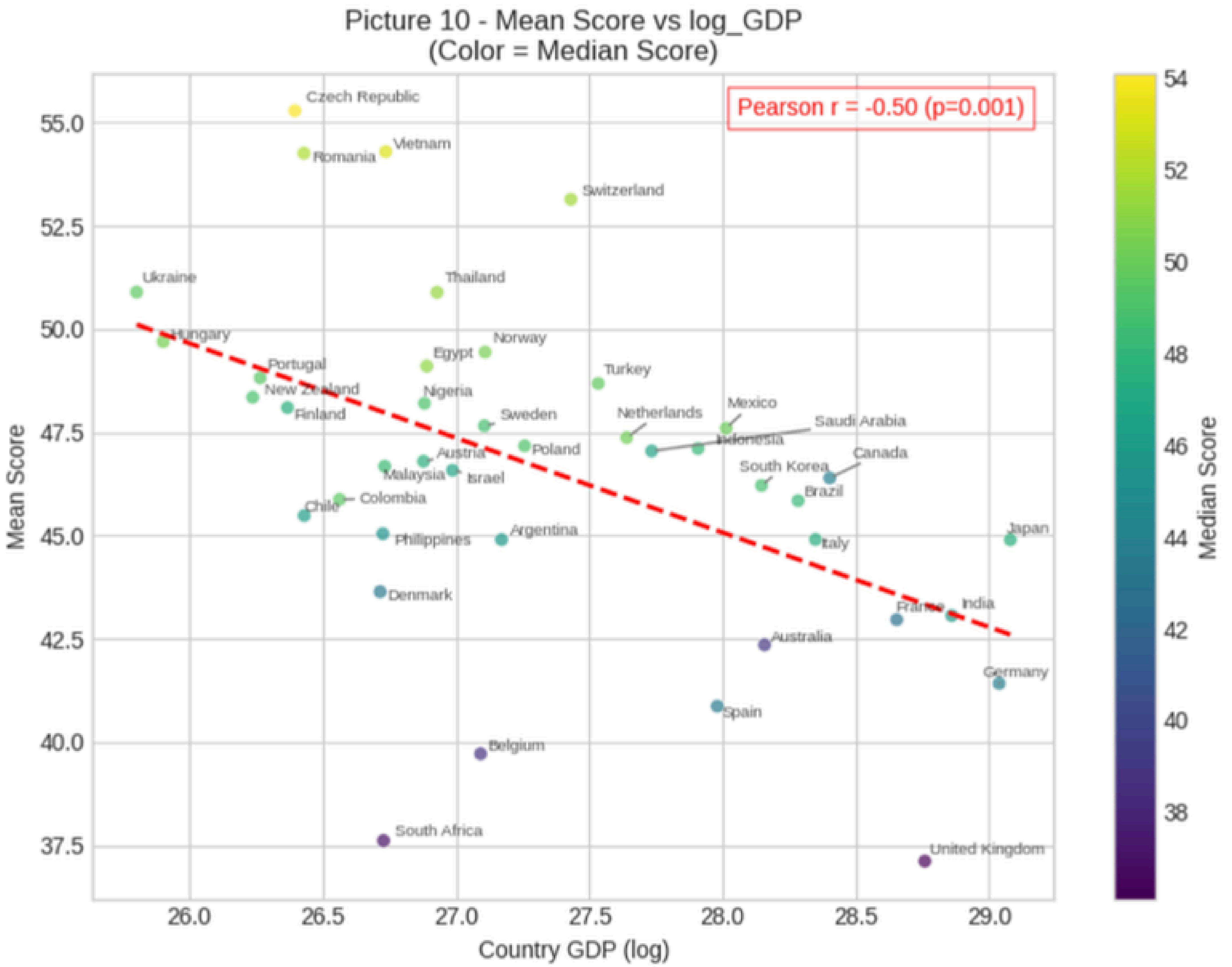
- **Sports** terms dominate global attention, even among longer-lived Rising terms.

- Global search volatility reflects not just trending terms but each region’s structural content mix.
- These category biases explain why some regions show more event-driven spikes, while others remain more stable or show recurring spam patterns.

Category Mix of Unique Terms

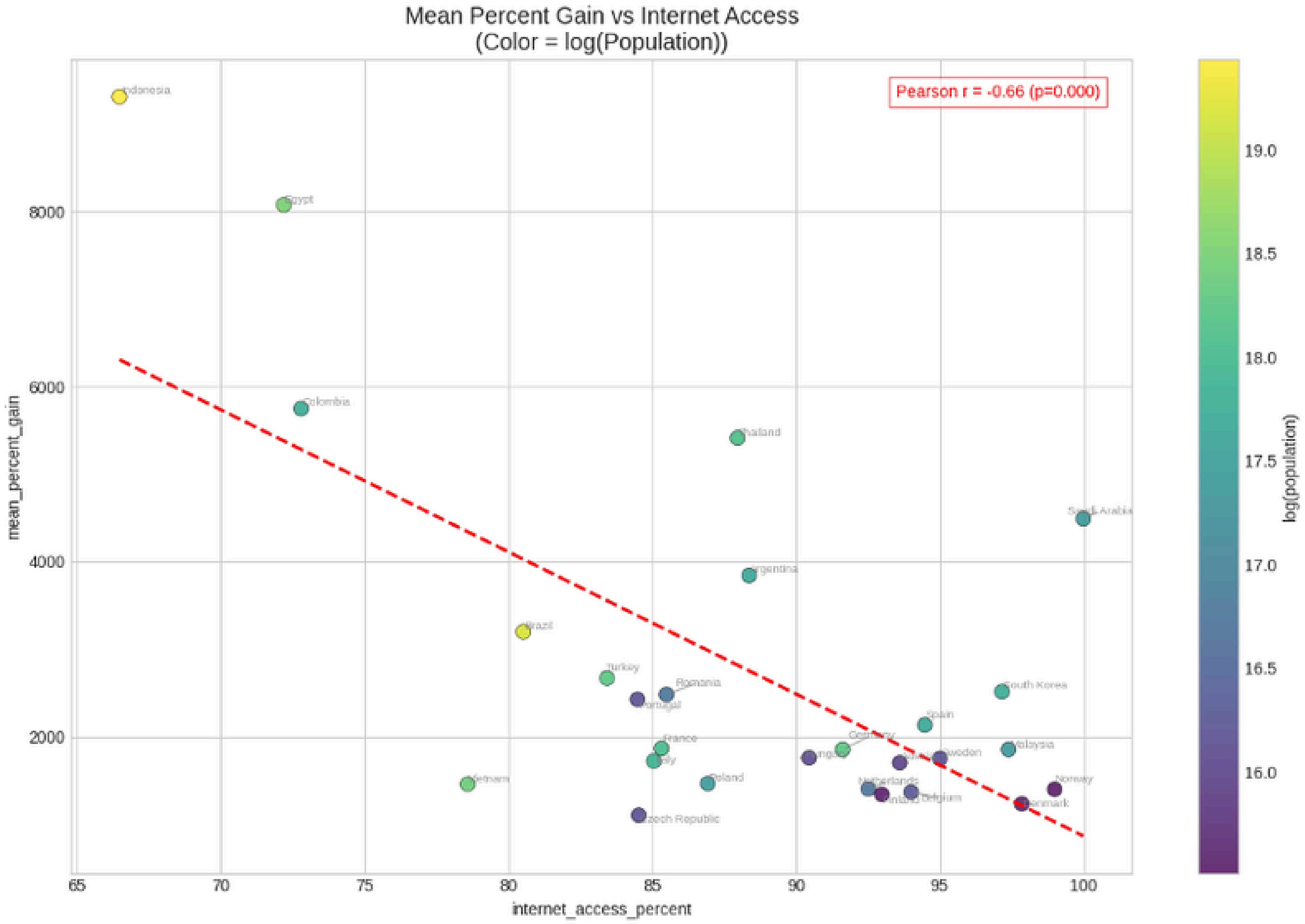
- Sports (matches and leagues): **45.7%**
- Entertainment (celebrities and culture): **15.7%**
- News and current events: **13.6%**
- General information and utilities: **9.4%**
- Gambling and betting terms: **8.4%**
- Finance and tech-related terms: **7.3%**

Country GDP as a Predictive Signal



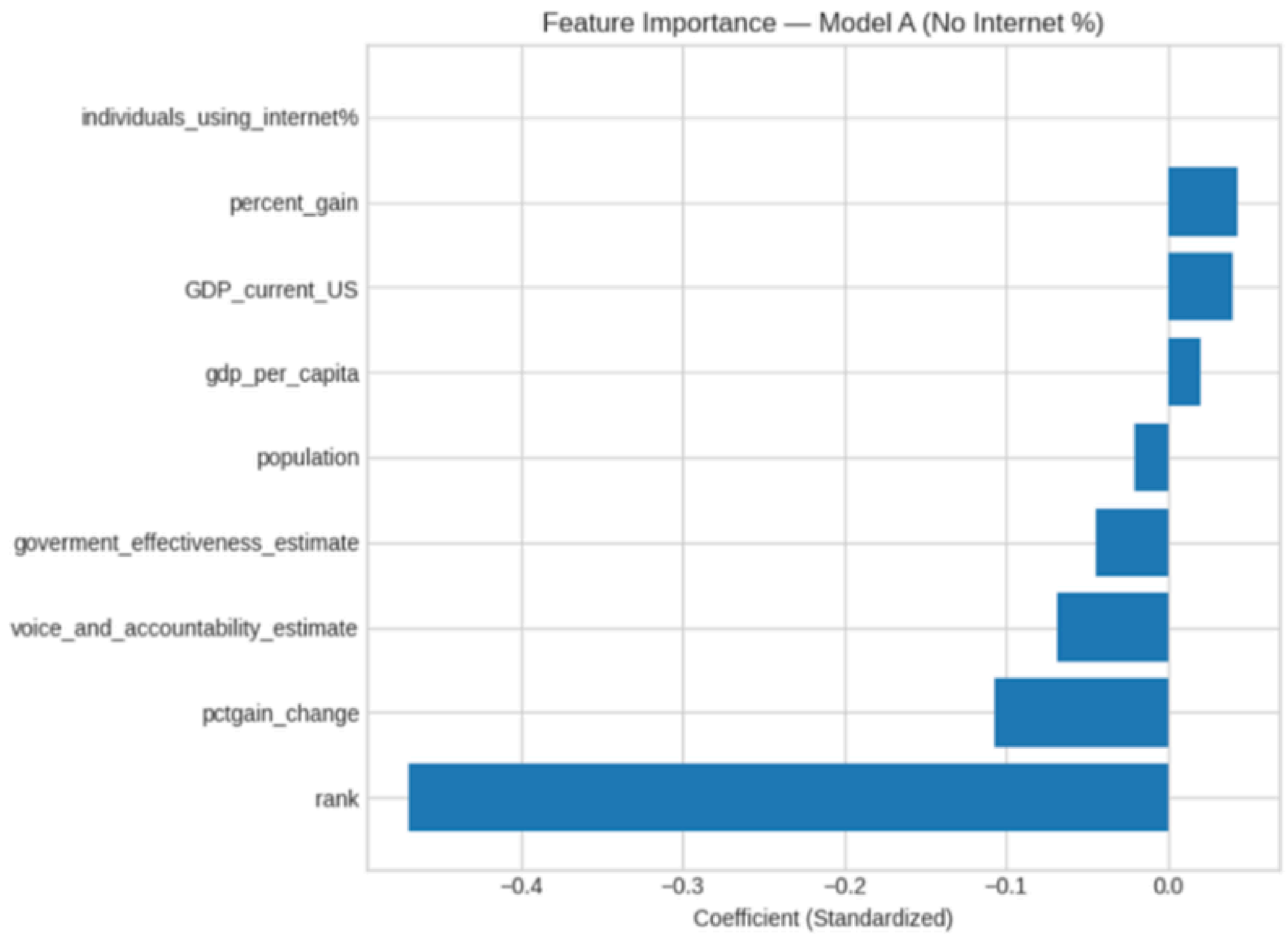
The **downward-sloping regression** line and a **Pearson correlation** of approximately **−0.50** indicate a moderately strong negative relationship: as **GDP rises**, **attention intensity falls**. The medium score also shows certain trend. From here, We can see that **Smaller economies** like Ukraine and Vietnam focus more on **dominant topics,** while **larger ones** like Japan and Germany **have more scattered and diversed attention**.

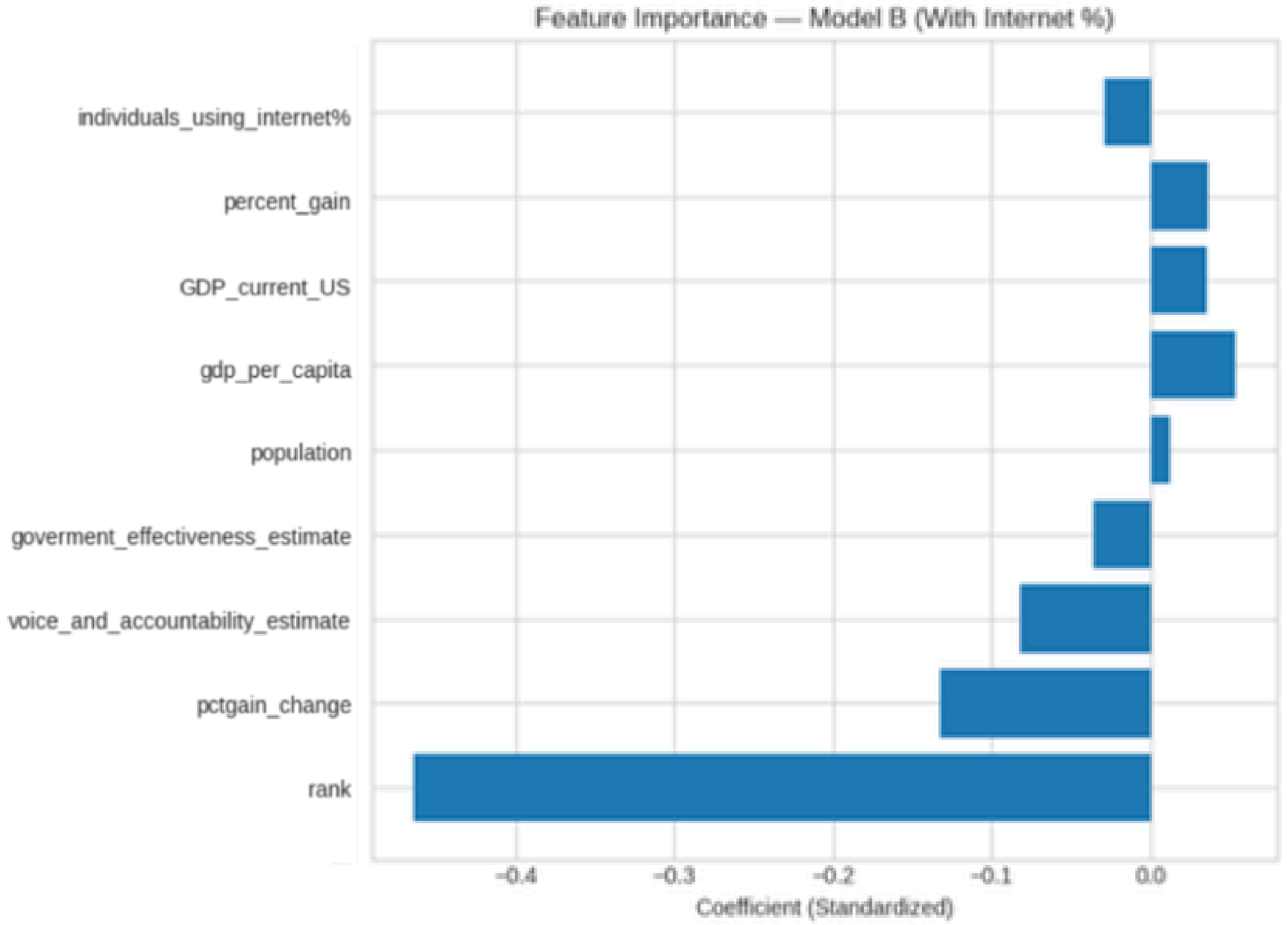
Digital Connectivity/Population as a Market Volatility Signal



The graph highlights **internet penetration** as a **key factor**, showing a strong **negative relationship (r = –0.66)**. Countries with **low internet access**, like Indonesia and Egypt, experience **high gains**, while places like Denmark see minimal spikes. **Digital maturity**, **population can only add as covariance**, drives **volatility**; so, we can see that **less-connected markets** see dramatic shifts as **connectivity grows**.

ML Model Architecture





Task 01

Classify whether a rising term in a country is **"sustainable"** — meaning whether it continues to appear in the next Google Trends rising-terms refresh cycle.

Model 02

We used a simple, interpretable **classification** approach to test whether trends show predictable **patterns of continuation** based on how they behaved in previous refresh cycles.

features 03

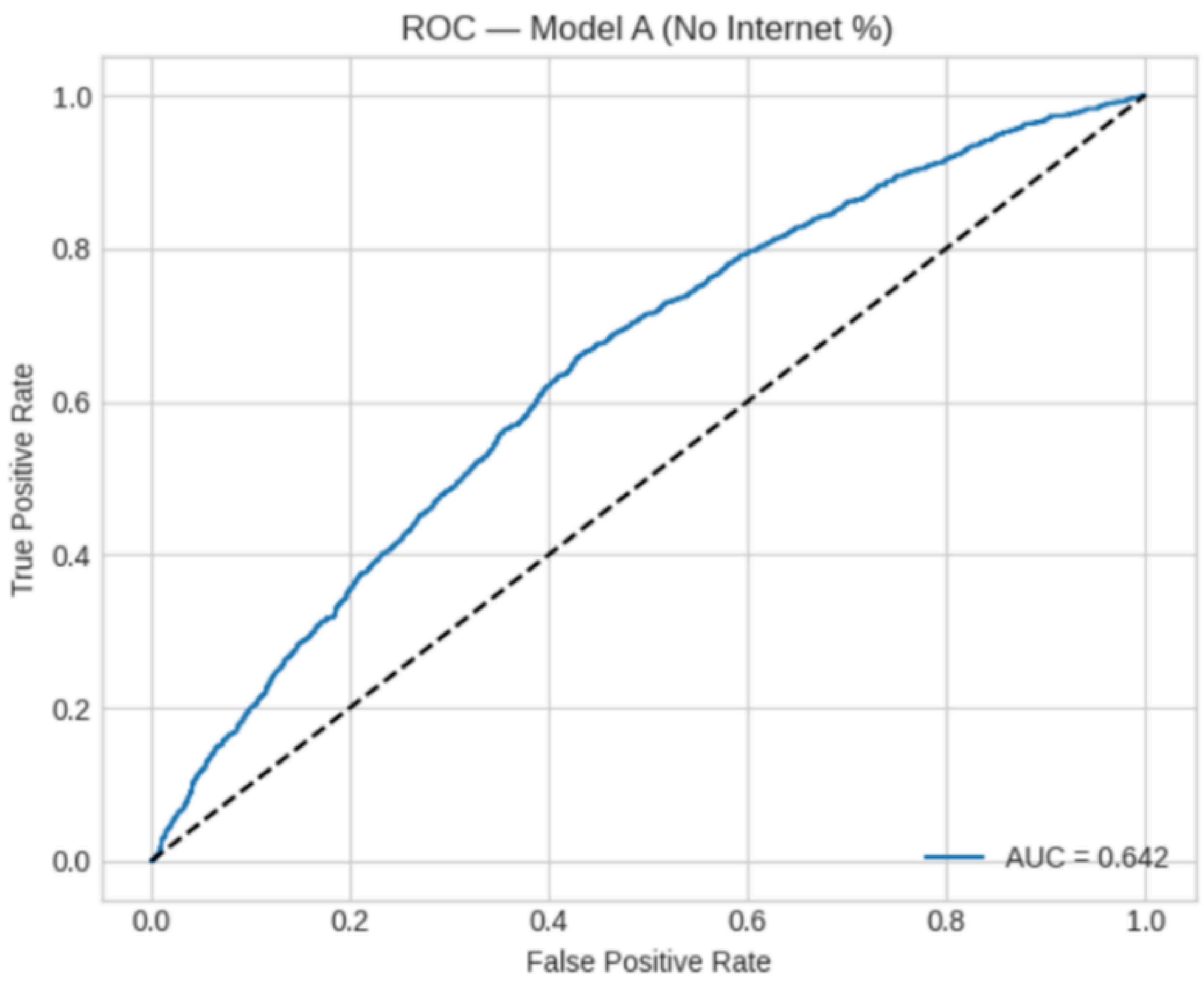
Our features captured both **trend dynamics** (percent gain, volatility) and **country characteristics** (GDP, governance) to see what shapes sustained attention.

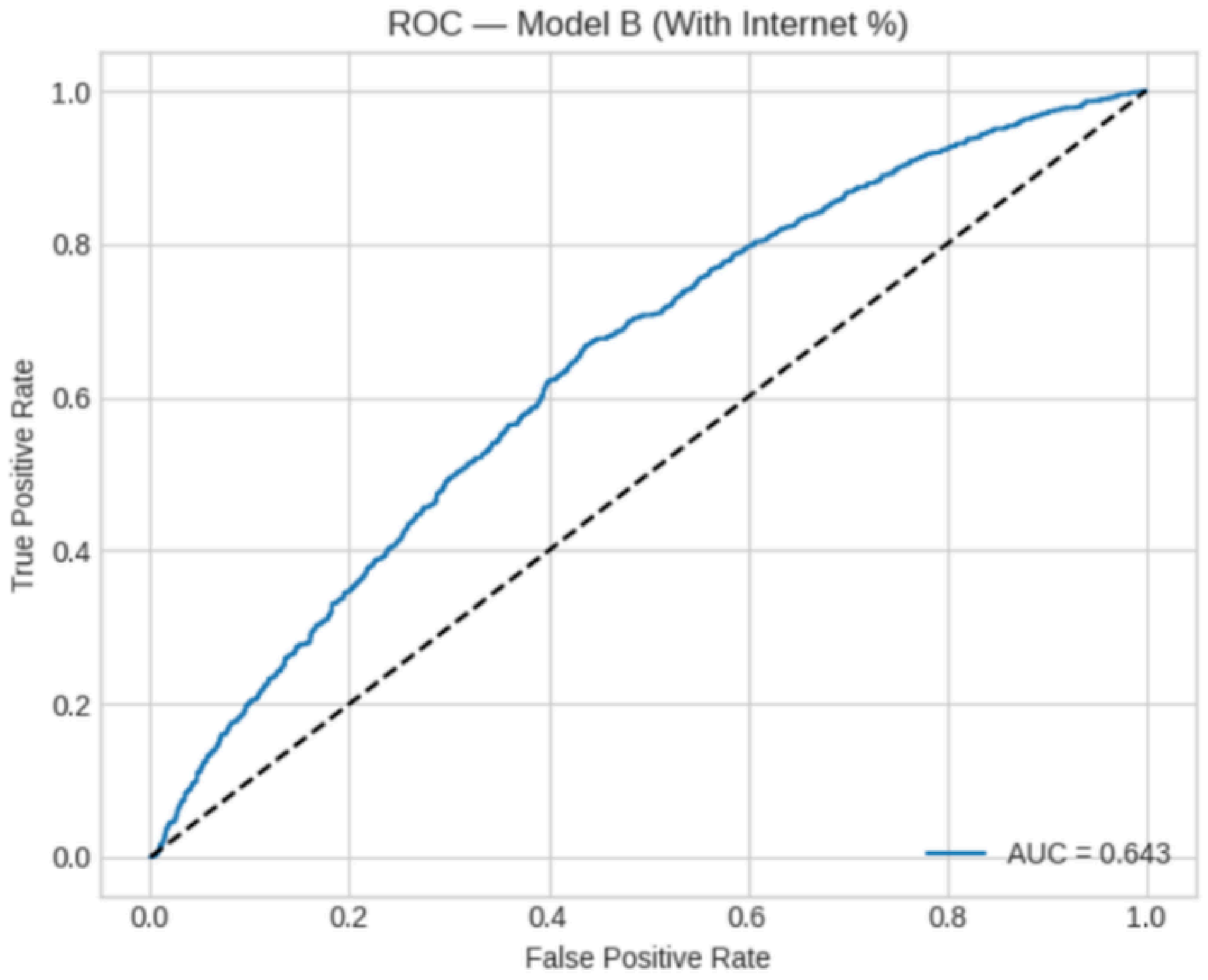
Comparision 04

Because internet-usage data is missing for many countries, we built two versions:

Model A: excludes internet %, keeps all countries.

Model B: includes internet %, but uses fewer countries.





What the Model Predicts

Performance

Model A and B perform almost identically **(AUC ≈ 0.64)**, showing that our features capture modest but meaningful signal.

Feature Directionality

Adding Internet % barely changes performance, suggesting that continuation is driven more by trend behavior than by country-level internet access.

Implication

Trend continuation is inherently noisy; the model is useful for diagnostics and understanding drivers, but not for high-accuracy forecasting.

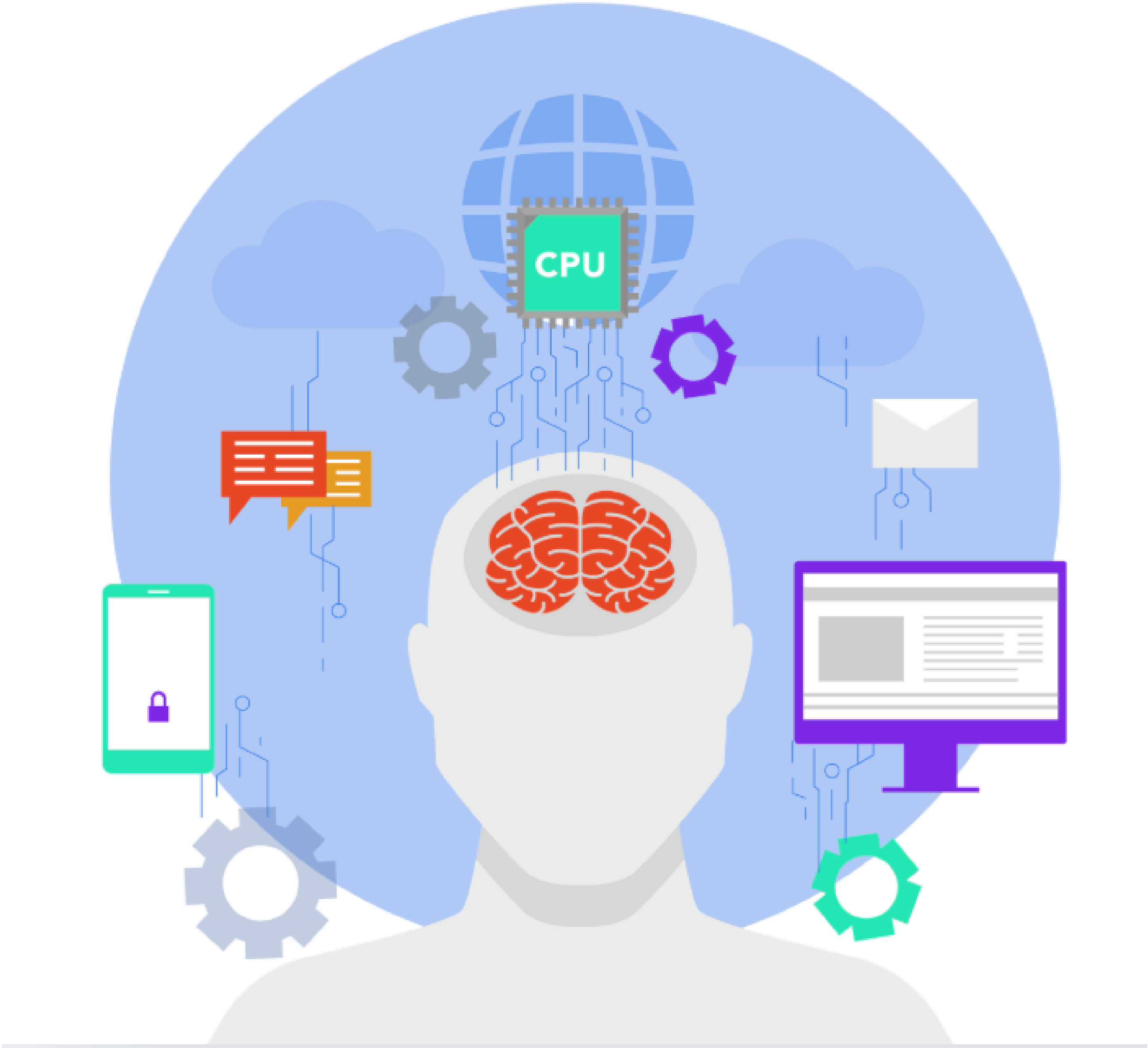
Conclusion and Takeaways

Rising Terms exhibit **longer survival times** than Top Terms, contradicting our initial assumption

Countries vary in **search attention**; some focus on sports, others on diverse content.

Not All Global Trends Are Organic.
Coordinated **Spam Exists**

Lower GDP nations exhibit **concentrated** search **attention**, unlike higher-GDP countries' scattered patterns.



Rising-term percent gain is **highest in emerging economies** like Indonesia, Egypt, and India, and lowest in developed ones.


THANK YOU G